COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	8/15/26	**DATE SUBMITTED:**	8/15/26
NAME:	Reyes, Cyrill Joferson B	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 6.1 **Linear and binary search**

1. PROCEDURE OUTPUT

Answer the following questions:

1. What is a search tree in data structures?
2. What are the Different types of search algorithm in data structures? Differentiate each type of search.
3. What operations / implementations can be performed using binary and linear search operations?
4. What are the advantages in using binary search tree as data structure?
5. Give an example program using binary search and Linear search.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. What is a search tree in data structures?
A search tree is a tree data structure used to store and organize data so that values can be searched efficiently. Each node contains a value, and the values are arranged according to a specific rule.

2. What are the Different types of search algorithm in data structures? Differentiate each type of search.
Linear Search checks each element one by one and can be used on unsorted data, while Binary Search divides the data into halves and requires the data to be sorted, making it faster for large datasets.

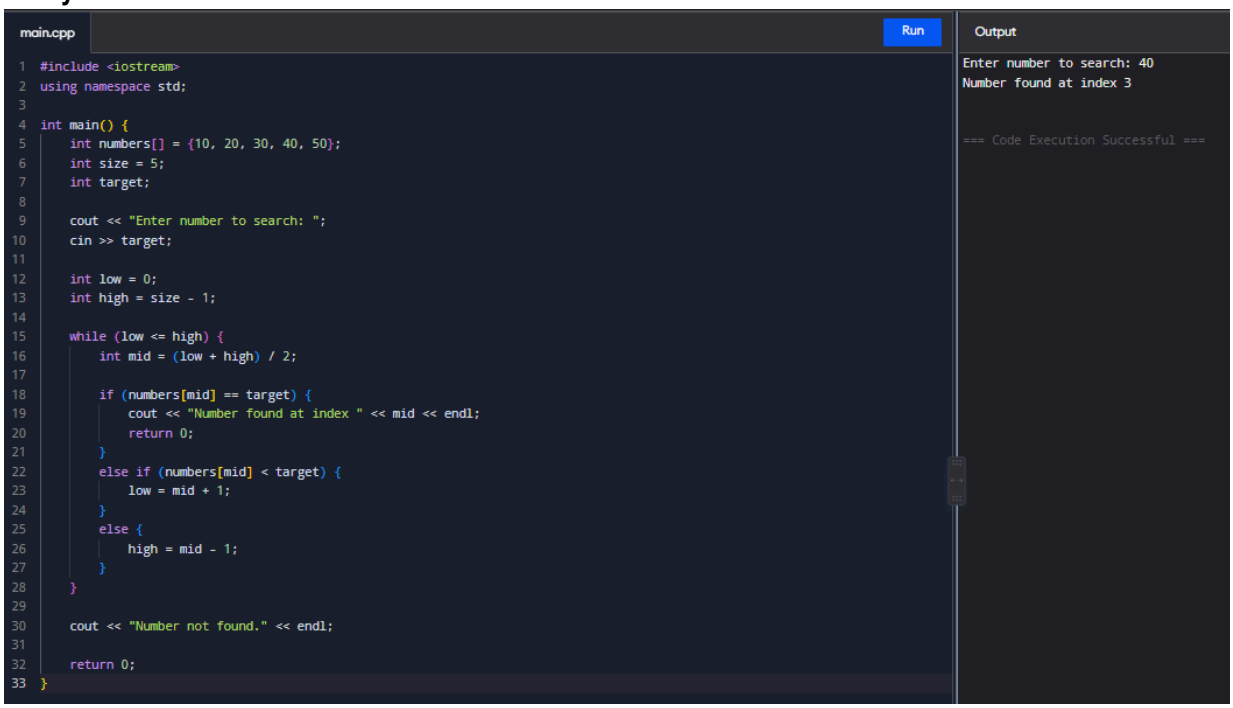
3. What operations / implementations can be performed using binary and linear search operations?
Both binary and linear search can be used to find an element, determine its position, and check whether an element exists in a collection of data.

4. What are the advantages of using a Binary Search Tree?
A Binary Search Tree provides efficient searching, insertion, and deletion, keeps data organized, and can quickly find values when the tree is properly balanced.

5. Example programs using Binary Search and Linear Search
Linear Search:



Binary Search:



3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

COMPUTER ENGINEERING

QUEZON CITY

Purpose of Use: **Grammar correction, idea generation, explanation of concepts, and code assistance.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was used to help improve the grammar, provide explanations, and assist with understanding and writing the answers. The final answers were reviewed and edited based on my own understanding.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.